

Discrete Mathematics - Practical 7

Set partitions and inclusion-exclusion: worked solutions
with the reasoning spelled out, step by step

Warm-up

Two tools run through this sheet. The first is the **Stirling number of the second kind** $S(n, k)$: the number of ways to split an n -element set into k non-empty unlabelled blocks. The second is the **inclusion-exclusion principle (IEP)**: to count elements that avoid a list of "bad" properties, start from the total, subtract each bad set, add back each pairwise overlap (subtracted twice), subtract triple overlaps, and so on. Most of this sheet is IEP in disguise - counting numbers with divisibility restrictions, or integer solutions of an equation with upper bounds, is the same alternating-sum idea. We solve problems 1, 7, 10 and 12.

Problem 1 - Splitting a set into non-empty classes

Problem 1

In how many ways can a 5-element set be partitioned into 3 non-empty classes? (The classes are unlabelled: only the grouping matters, not any names or order of the classes.)

The idea

This count is exactly the Stirling number of the second kind $S(5, 3)$, and the cleanest way to pin it down is a **recurrence built by focusing on one element**. Single out one fixed element, say the last one, x_5 . In any partition into 3 non-empty classes, exactly one of two things happens to x_5 :

- x_5 **is alone** in its own class. Then the other 4 elements must form the remaining 2 non-empty classes: $S(4, 2)$ ways.
- x_5 **joins others**. Then the other 4 elements already form all 3 non-empty classes ($S(4, 3)$ ways), and x_5 is dropped into one of those 3 classes: $3 \cdot S(4, 3)$ ways.

These two cases are mutually exclusive and cover everything, so we add them. This is the general recurrence $S(n, k) = S(n-1, k-1) + k S(n-1, k)$, and it lets us climb up from tiny known values.

Step 1 - Get the smaller values $S(4, 2)$ and $S(4, 3)$. Using the same recurrence (and $S(n, 1) = 1$, $S(n, n) = 1$):

$$S(4, 2) = S(3, 1) + 2 S(3, 2) = 1 + 2 \cdot 3 = 7, \quad S(4, 3) = S(3, 2) + 3 S(3, 3) = 3 + 3 \cdot 1 = 6.$$

(Here $S(3, 2) = S(2, 1) + 2 S(2, 2) = 1 + 2 = 3$.)

Step 2 - Combine.

$$S(5, 3) = S(4, 2) + 3 S(4, 3) = 7 + 3 \cdot 6 = 7 + 18 = 25.$$

Always check. Independent formula $S(n, k) = \frac{1}{k!} \sum_{j=0}^k (-1)^j \binom{k}{j} (k-j)^n$ gives

$$S(5, 3) = \frac{1}{6} (3^5 - 3 \cdot 2^5 + 3 \cdot 1^5) = \frac{243 - 96 + 3}{6} = \frac{150}{6} = 25.$$

A direct computer enumeration of all set-partitions of $\{1, 2, 3, 4, 5\}$ finds exactly 25 with three blocks. ✓

Answer: $S(5, 3) = 25$

★ Method to remember

Transferable technique - the Stirling recurrence.

$$S(n, k) = \underbrace{S(n-1, k-1)}_{\text{new element alone}} + \underbrace{k S(n-1, k)}_{\text{new element joins one of the } k \text{ blocks}}.$$

Singling out one element and asking "is it alone or does it join?" is a go-to move for counting structures recursively. **Beware:** if the classes were *labelled* (distinguishable boxes), the count would instead be $k! S(n, k)$, or, allowing empty boxes, k^n .

Building the table. Each entry is "upper-left neighbour" plus $k \times$ "left neighbour"; our target $S(5, 3) = 25$ is highlighted.

$n \setminus k$	1	2	3	4	5
1	1				
2	1	1			
3	1	3	1		
4	1	7	6	1	
5	1	15	25	10	1

Problem 7 - Divisible by 11, but by neither 5 nor 3

Problem 7

Count the integers from 1 to 16500 that are divisible by 11 but are **not** divisible by 5 and **not** divisible by 3.

The idea

The phrase "divisible by 11" fixes our universe: forget every other number and look only at the 1500 multiples of 11 in range. Inside that universe we must throw out the ones that are *also* divisible by 5 or by 3. "Or" is the signal for **inclusion-exclusion**: if we just add (multiples of 5) and (multiples of 3), the numbers divisible by *both* 5 and 3 get counted twice, so we add them back once.

A clean trick keeps the arithmetic exact: "multiple of 11 and multiple of 5" means "multiple of 55" (since 5, 11 are coprime), and similarly 3 gives 33, and 5 and 3 together give 165. Because $16500 = 11 \cdot 1500$ and 16500 is also a multiple of 55, 33 and 165, every floor below is an exact division - no rounding to worry about.

Step 1 - The universe: multiples of 11.

$$\left\lfloor \frac{16500}{11} \right\rfloor = 1500.$$

Step 2 - The "bad" ones (also divisible by 5 or 3). Within the multiples of 11:

$$\#(\text{also } 5) = \frac{16500}{55} = 300, \quad \#(\text{also } 3) = \frac{16500}{33} = 500, \quad \#(\text{also } 5 \text{ and } 3) = \frac{16500}{165} = 100.$$

Step 3 - Inclusion-exclusion for "or". Multiples of 11 that are divisible by 5 **or** 3:

$$300 + 500 - 100 = 700.$$

Step 4 - Subtract from the universe.

$$1500 - 700 = 800.$$

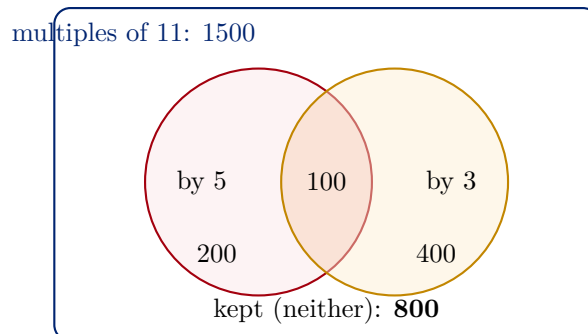
Always check. A brute-force loop over $1, \dots, 16500$ counting n with $11 \mid n$, $5 \nmid n$, $3 \nmid n$ returns exactly 800. Cross-check by fractions: among multiples of 11, the fraction surviving "not 5, not 3" is $\frac{4}{5} \cdot \frac{2}{3} = \frac{8}{15}$, and $1500 \cdot \frac{8}{15} = 800$ (this clean fraction works because 1500 is a multiple of 15). ✓

Answer: $1500 - (300 + 500 - 100) = 800$

★ Method to remember

Transferable technique - count in the right universe, then IEP. For " A but not B or C ": restrict to A , then subtract $|B \cup C| = |B| + |C| - |B \cap C|$ (all measured inside A). For divisibility, "multiple of a and of b " means "multiple of $\text{lcm}(a, b)$ ", and the count up to N is $\lfloor N/\text{lcm} \rfloor$.

Venn view (counts inside the 1500 multiples of 11). We delete the shaded overlap region $300 + 500 - 100 = 700$ and keep the 800 outside.



Problem 10 - Triangles in a side-8 subdivided triangle

Problem 10

Each side of triangle ABC is divided into 8 equal parts, and through the division points all lines parallel to the three sides are drawn, producing a triangular grid. Count all triangles whose vertices are grid points (the corners A, B, C count as grid points) and whose three sides are each parallel to a side of ABC . This includes triangles in the same orientation as ABC (**upward**) and the inverted ones (**downward**).

The idea

"Sides parallel to ABC " means every triangle we count is similar to ABC and sits squarely on the grid; there are only two possible orientations, **upward** (pointing the same way as ABC) and **downward** (inverted). The grid has small upward unit triangles and small downward unit triangles, and a bigger triangle of either orientation is just a union of small ones. So the plan is: count upward triangles of every size, count downward triangles of every size, and add.

Why split by orientation? Because the two families obey *different* size limits. An upward triangle of side k fits as long as $k \leq n$. But a downward triangle of side k needs room on all sides; it only fits when k is at most $\lfloor n/2 \rfloor$, which is why large inverted triangles simply do not exist. Keeping the two counts separate is what makes the problem manageable. Here $n = 8$.

Step 1 - Upward triangles of side k . An upward triangle of side k is fixed by the position of its top vertex. That vertex can sit anywhere that still leaves room for the side- k triangle below it, and those positions form a triangular array of side $n - k$, which holds $1 + 2 + \dots + (n - k + 1) = \frac{(n-k+1)(n-k+2)}{2}$ points. Each size count is itself a binomial, $\frac{(n-k+1)(n-k+2)}{2} = \binom{n-k+2}{2}$, so summing over all sizes $k = 1, \dots, 8$ is $\binom{2}{2} + \binom{3}{2} + \dots + \binom{9}{2}$. By the hockey-stick identity ($\sum_{m=2}^{n+1} \binom{m}{2} = \binom{n+2}{3}$) this collapses to a single binomial coefficient:

$$\sum_{k=1}^8 \frac{(n-k+1)(n-k+2)}{2} = \binom{n+2}{3} = \binom{10}{3} = 120.$$

Term by term: 36, 28, 21, 15, 10, 6, 3, 1 for $k = 1, \dots, 8$, and $36 + 28 + 21 + 15 + 10 + 6 + 3 + 1 = 120$.

Step 2 - Downward (inverted) triangles of side k . A downward triangle of side k exists only for $k \leq \lfloor 8/2 \rfloor = 4$, and there are $\frac{(n-2k+1)(n-2k+2)}{2}$ of side k :

$$k = 1 : \frac{7 \cdot 8}{2} = 28, \quad k = 2 : \frac{5 \cdot 6}{2} = 15, \quad k = 3 : \frac{3 \cdot 4}{2} = 6, \quad k = 4 : \frac{1 \cdot 2}{2} = 1.$$

Sum of downward triangles: $28 + 15 + 6 + 1 = 50$.

Step 3 - Total.

$$\underbrace{120}_{\text{upward}} + \underbrace{50}_{\text{downward}} = 170.$$

Always check. A direct computer enumeration over the triangular grid (test every triple of grid points forming an upward or a downward triangle of any size) returns 120 upward and 50 downward, total 170. A known closed form for a side- n triangle also agrees: for $n = 8$, $\lfloor \frac{n(n+2)(2n+1)}{8} \rfloor = \frac{8 \cdot 10 \cdot 17}{8} = 170$. ✓

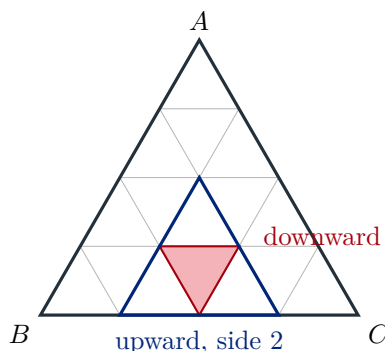
Answer: 120 (upward) + 50 (downward) = 170

Important reading note. If the intended count were only triangles *similarly oriented* to ABC (upward), the answer would be 120. The statement asks for all grid triangles with sides parallel to ABC , which includes the inverted ones, so the full total is **170**. We report the breakdown so either reading is covered.

★ Method to remember

Transferable technique - count by size and by orientation. In a triangular grid of side n : upward triangles total $\binom{n+2}{3}$ (sum of $\frac{(n-k+1)(n-k+2)}{2}$). Downward triangles of side k exist only for $2k \leq n$ and number $\frac{(n-2k+1)(n-2k+2)}{2}$. When a figure can appear in more than one orientation, split the count by orientation first - the orientations usually have different feasibility limits.

The grid (here drawn at side 4 for clarity). Upward triangles point up like ABC ; the one shaded triangle is *downward*. A downward triangle of side k fits only when $2k \leq n$ (it must sit inside an upward triangle of side $2k$), which is why the big inverted ones run out of room.



Counts by size for $n = 8$ (what we actually summed):

side k	1	2	3	4	5	6	7	8	total
upward	36	28	21	15	10	6	3	1	120
downward	28	15	6	1	0	0	0	0	50

Problem 12 - Bounded integer solutions of a linear equation

Problem 12

Find the number of integer solutions of the system

$$x_1 + x_2 + x_3 = 40, \quad 4 \leq x_1 \leq 15, \quad 9 \leq x_2 \leq 18, \quad 5 \leq x_3 \leq 16.$$

The idea

Two obstacles stand between us and a plain stars-and-bars count: the variables have **lower** bounds (they do not start at 0) and **upper** bounds (they cannot grow without limit). We remove them one at a time.

Lower bounds are free: substitute $y_i = x_i - (\text{its minimum})$, so each $y_i \geq 0$ and the total drops accordingly. **Upper bounds need inclusion-exclusion:** first count solutions ignoring the upper caps (easy stars and bars), then subtract the "illegal" solutions where some y_i overshoots its cap. The clever part: " y_i exceeds its cap c_i " means " $y_i \geq c_i + 1$ ", which we enforce by handing $c_i + 1$ units to y_i up front and counting the rest freely. Overshoots of two variables at once get subtracted twice, so IEP adds them back. The whole solution is one tidy alternating sum.

Step 1 - Kill the lower bounds. Let $y_1 = x_1 - 4$, $y_2 = x_2 - 9$, $y_3 = x_3 - 5$. Then $y_i \geq 0$ and

$$y_1 + y_2 + y_3 = 40 - 4 - 9 - 5 = 22, \quad 0 \leq y_1 \leq 11, \quad 0 \leq y_2 \leq 9, \quad 0 \leq y_3 \leq 11.$$

(The caps are $15 - 4 = 11$, $18 - 9 = 9$, $16 - 5 = 11$.)

Step 2 - Count ignoring the upper caps. Nonnegative solutions of $y_1 + y_2 + y_3 = 22$ by stars and bars:

$$\binom{22 + 3 - 1}{3 - 1} = \binom{24}{2} = 276.$$

Step 3 - Subtract single overshoots. " y_i over its cap" means $y_i \geq \text{cap} + 1$; give it that many units and count solutions of the reduced total $22 - (\text{cap} + 1)$:

$$|y_1 \geq 12| = \binom{(22 - 12) + 2}{2} = \binom{12}{2} = 66, \quad |y_2 \geq 10| = \binom{(22 - 10) + 2}{2} = \binom{14}{2} = 91,$$

$$|y_3 \geq 12| = \binom{(22-12)+2}{2} = \binom{12}{2} = 66.$$

Step 4 - Add back double overshoots. Two caps exceeded at once (give both their cap + 1 units, then count the rest):

$$|y_1 \geq 12, y_2 \geq 10| = \binom{(22-12-10)+2}{2} = \binom{2}{2} = 1, \quad |y_2 \geq 10, y_3 \geq 12| = \binom{2}{2} = 1,$$

$$|y_1 \geq 12, y_3 \geq 12| : 22 - 12 - 12 = -2 < 0 \Rightarrow 0.$$

All three caps at once needs $12 + 10 + 12 = 34 > 22$ units, impossible, so the triple term is 0.

Step 5 - Assemble (inclusion-exclusion).

$$276 - (66 + 91 + 66) + (1 + 1 + 0) - 0 = 276 - 223 + 2 = 55.$$

Always check. A brute-force triple loop over $x_1 \in [4, 15]$, $x_2 \in [9, 18]$, $x_3 \in [5, 16]$ counting those with $x_1 + x_2 + x_3 = 40$ returns exactly 55. ✓

$$\text{Answer: } \binom{24}{2} - \left[\binom{12}{2} + \binom{14}{2} + \binom{12}{2} \right] + [1 + 1] = 55$$

★ Method to remember

Transferable technique - bounded compositions. To count integer solutions of $x_1 + \dots + x_k = N$ with $a_i \leq x_i \leq b_i$:

1. Substitute $y_i = x_i - a_i$ to make all lower bounds 0 (and lower N by $\sum a_i$).
2. Count without upper caps: $\binom{N'+k-1}{k-1}$.
3. By IEP, subtract/add terms where a chosen subset S of variables overshoots: each such variable contributes $(\text{cap}_i + 1)$ removed from the total, with sign $(-1)^{|S|}$. Drop any term whose remaining total goes negative (it contributes 0).

The IEP bookkeeping at a glance.

overshoot set	remaining total	$\binom{+2}{2}$	sign	contribution
none	22	276	+	+276
$y_1 \geq 12$	10	66	−	−66
$y_2 \geq 10$	12	91	−	−91
$y_3 \geq 12$	10	66	−	−66
$y_1 \geq 12, y_2 \geq 10$	0	1	+	+1
$y_2 \geq 10, y_3 \geq 12$	0	1	+	+1
$y_1 \geq 12, y_3 \geq 12$	−2	0	+	0
all three	−12	0	−	0
net total (sum of contributions)				55